
UNIT 21 MACRO-VARIABLE DATA: NATIONAL INCOME SAVING AND INVESTMENT

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21.0 OBJECTIVES

After going through this Unit, you will be able to:

- know the various approaches followed by the Indian statistical system in generating data on various economic and social phenomena;
- explain the SNA (System of National Accounts) methodology of data compilation;
- identify the various aggregates on which estimates are made by the National Accounts System;
- compare the interrelationship among different sectors of the economy through input-output transaction table (I-OTT) compiled by CSO; and
- describe the data on National Income, Saving, Investment and other related macro economic variable and the estimates of the state income which enable analysis of various aspects of economy.

21.1 INTRODUCTION

You would soon be a professional economist and would possibly be engaged in research or would function as an economic analyst or adviser in a Government or non-Government organization. As a professional economist, you will have to analyse situations and arrive at conclusions about them or develop solutions

for further action to tackle such situations. You will, for this purpose, identify the variables that, in your opinion, shape or affect the problem at hand, incorporate these in a suitably chosen model, test the efficacy of the model in reflecting the situation under examination and proceed further with the analysis. A basic input for such activity is data on the variables incorporated in the model. The availability of data **relevant** to the proposed analysis enhances or restricts the scope of your analysis and the reliability of the conclusions arrived at on the basis of the analysis. Indeed, it might even restrict the kind of variables and model you would like to use in your analysis, unless you want to conduct a survey yourself to collect the data that are not available. Your technical and professional equipment as an economist is, therefore, incomplete without knowledge of where data on the variables in question are available and how good they are for the purpose of the analysis at hand. This knowledge should naturally cover the entire data base of the Indian economy since an economist's work can cover any part or the whole of the Indian economy.

We shall look at the data base of the Indian economy in this Block. This Unit deals with data available on National Income, Saving and Investment – the major macro variables relating to the economy and associated macroeconomic aggregates. Agricultural and Industrial data, data relating to the Social Sector, Trade and Finance will be discussed in the subsequent units of this block.

Let us begin by taking a look at the Indian statistical system.

21.2 THE INDIAN STATISTICAL SYSTEM

The Indian Statistical System generates data on a variety of economic and social phenomena, essentially through six approaches. First, Central Acts like the Census Act, 1948; and the Collection of Statistics Act, 2008, etc. enable the Government agencies to conduct large-scale sample surveys for collection of data at regular intervals. Second, statutory returns prescribed under several other Acts like the Factories Act, 1948 the Companies Act, 1956, the Reserve Bank of India Act, 1953; the Registration of Births and Deaths Act, 1969 and the implementation of these Acts generate data on matters not covered by the surveys. Third, data collected by individual Ministries, Departments and organizations of the Central and State Governments as part of their specific functions reflect the emerging situation in different sectors and sub-sectors of the economy and administrative divisions of the country. Fourth, the administrative reports of these organizations supplement such data. Next, information derived from the data flows mentioned above, like the **National Accounts Statistics (NAS)**, index numbers of prices and production and indices like the Human Development Index and Gender Development Index provide readily usable inputs for research and policy and evaluation of the impact of policies and programmes of development in terms of the health of the economy and the well-being of the society. Lastly, a large number of surveys and research studies conducted by various institutions, public and private, on a variety of subjects constitute another flow of data and information.

The Ministry of Statistics and Programme Implementation of the Government of India is the apex body in the official statistical system of the country. The Ministry is headed by the Chief Statistician of the country. The Ministry consists of the Central Statistics Office (CSO), the National Sample Survey Office (NSSO) and the Computer Centre (CC). The Directorates of Economics and Statistics function at the level of the State/UT Governments. The CSO

coordinates the statistical activities in the country, lays down and maintains statistical norms and standards and provides liaison with Central, State and International statistical agencies. There is also a National Statistical Commission (NSC), comprising of Chairman and Members, who are eminent economists and statisticians from research institutions, representatives of Central Ministries and Departments and the State Directorates of Economics and Statistics, to provide (i) guidance for an overall perspective for statistical development in the country, (ii) guidance to Government on policy issues, and (iii) ensure effective coordination of all statistical activities of the Government of India. CSO is the Secretariat of the NSC.

The CSO, as the Central Statistical Authority, is responsible for coordination of statistical activities in the country and for evolving and maintaining statistical standards. Mention should be made in this connection of three publications of CSO, namely, the Sources and Methods of National Accounts Statistics, the National Industrial Classification and the Consumer Price Index. The first two are *ad hoc* while the third a monthly one. The first one covers the methods and classification principles to be followed for preparation of the macro economic aggregates, the second one gives the classification principle to be followed for national and international comparability and the third one provides State/UT wise price indices.

Certain other agencies concerned with economic development also bring out publications that cover data relating to most of all the sectors of the economy at one place. These are **the Reserve Bank of India (RBI) Bulletin (monthly), the Financial Stability Report (including Trend and Progress of Banking in India), RBI's Handbook of Statistics on the Indian Economy and the website of the RBI**, the pre-Budget Economic Survey and the **Budget presented to Parliament** by the Finance Minister every year. **The Five-Year Plan documents of the Planning Commission of the Government of India used to make up yet another set of data sources that provides important data across sectors. The Human Development Report** was prepared by the Planning Commission **for the first time in the year 2001. This and similar reports prepared subsequently by many of the State Governments** for individual States and for district constituted another set of sources for data across sectors at the national, State, district and even sub-district levels. The **District Census Handbooks** published after successive population censuses by the Office of the **Registrar General of India (ORGI) and the Directorates of Census of different States and Union Territories** provide comprehensive information across sectors on individual districts, sub-districts, towns and villages. The ORGI, in the year 2014 has also launched a digital library of all census tables published since Census 1991 in their website www.censusindia.gov.in. These tables together provide a very useful time series data on demographic characteristics of the country and its States/ UTs. Most of the data and reports published by Government agencies are available in electronic format, that is, in DVDs/CDs and are accessible at the websites of the agencies concerned. In the year 2013, the Planning Commission has also set up a central web-based repository, namely, www.data.gov.in to bring published data of different Government agencies under one roof.

The Indian statistical system has also recently been reviewed so as to bring about improvements in it. A National Statistical Commission was appointed by the Government of India to examine critically the deficiencies of the existing statistical system from the point of view of timely availability, reliability and adequacy of data and to recommend measures to correct these deficiencies and

revamp the statistical system to generate reliable statistics for the purpose of policy and planning at different levels of Government. The Commission submitted its report to Government in September, 2001. **The Report of National Statistical Commission** (2001), published by the **Ministry of Statistics and Programme Implementation (MoSPI)**, is itself a useful reference book on the database of the Indian economy. This can be viewed or downloaded from the website www.mospi.nic.in/nscr/hp.htm. Most of data available in Government websites can be downloaded free of cost, either directly or as a registered user. In most cases, this registration is free of charge. For example, one can make a free registration in the website of the MoSPI and with the username, freely download all the reports and tables published by different offices under the MoSPI. For the Office of the RGI, one can simply visit its website www.censusindia.gov.in click on the link digital library and download all the census tables published from 1991 census onwards.

There are also non-Government sources that publish these for the use of researchers and other data users (Certain private websites like www.indiastat.com provide India centric, sector specific and state specific data for the research fraternity, usually for a fee). The **Centre for Monitoring Indian Economy (CMIE)**, Mumbai and its **Economic Intelligence Services (EIS)** provide detailed and up-to-date information on the Indian economy through a **Monthly Review of Indian Economy** and annual documents each covering a specific sector or subject in great detail. These publications bring together at one place not only data available with Government statistical agencies but also data collected by CMIE itself. The **Economic and Political Weekly Research Foundation (EPWRF)**, Mumbai is another organization that publishes detailed time series data on a variety of subjects along with a description of related conceptual and methodological issues and quality of data, besides presenting statistics on specific areas in **subject-specific Special Issues of the Economic and Political Weekly**. The search for data by an analyst like you can well start from one or more of the comprehensive sources enumerated above and move on, as necessary, to the primary agency concerned with the specific area of your interest.

Data on any variable have to be classified suitably for facilitating analysis. And the classification system needs to be standardized when data on any variable are collected by several agencies and utilized by many in order to ensure meaningful collection of data and comparability of data across space and time. At the international level, different agencies of the United Nations, like, the United Nations Statistics Division (UNSD), the OECD, the International Labour Office (ILO), etc. prepares these classifications for cross-country comparison. For individual countries, these classification systems are recommended. The countries are advised to maintain full compatibility up to a certain level and then make additions/ alterations to suit the country specific needs. In India, usually, such classifications are expanded to suit the need of the country, while maintaining international comparability. Two important classificatory systems developed for use may be referred to here, as its utility cuts across sectors. One is the National Industrial Classification (NIC), developed by the Central Statistics Office (CSO) in consultation with all concerned. It groups the entire spectrum of economic activities by means of a five-digit code structure. NIC has been revised from time to time in keeping with changes in the structure and variety of economic activities over the years. A Standard Industrial Classification, 1962 (SIC 62) developed by CSO was being used in the Sixties until NIC 70 was introduced. This was replaced by NIC 1987, which in turn was replaced by NIC 98 in December 1998. The most

recent revision of the NIC series is NIC 2008. All the NIC classifications are based on the relevant UN International Standard Industrial Classification (ISIC). NIC 2008 was, for instance, based on ISIC- Revision 4. NIC 2008 has been developed and released for use in the place of NIC 04 in September, 2008. NIC 2008 classifies the whole range of economic activity into 21 Sections, 88 Divisions, 238 Groups, 403 Classes and 1,304 Sub-classes. NIC 2008 is comparable completely with ISIC-Rev.4 up to the four-digit level. The fifth digit (sub-class) inside each of the class (4th digit) takes care of the special features of the Indian economy. NIC is used for classifying data on variables like output, employment and income by economic activity. In each release of the NIC, concordance tables are made available to facilitate recasting of past data as per the new classification, to the extent possible.

The other classificatory system is the **National Classification of Occupations (NCO)**, developed by the **Directorate General of Employment & Training (DGE&T) of the Ministry of Labour & Employment**. **NCO 1958**, a five-digit code structure, was developed in the Fifties. This was revised to **NCO 1968**. This was in use till recently. This has now been replaced by **NCO 2004 (NCO04)**. The Indian NCOs have been based on the international occupational classification system developed by the International Labour Organisation (ILO) and revised from time to time. NCO 1968 was patterned on the **International Standard Classification of Occupations 1966 – ISCO 66 – of the ILO**. While revising ISCO 66 to prepare **ISCO 88**, ILO incorporated the concept of skill for effective performance in the relevant occupation, utilizing for the purpose the **International Standard Classification of Education (ISCED)**. **NCO 04 has followed the approach adopted in ISCO 88**. It takes note of the skill required for satisfactory performance in the relevant occupations and has a six-digit code structure. NCO 04 has 8 occupational divisions (the first digit of the code), 95 occupational groups (the second digit), 462 occupational families (the third digit) and 2484 occupations (the fifth and the sixth digits). NCO is useful for classification of the job seekers, the jobs and employment, thereby facilitating analysis of trends in the labour market and in the employment potential of the economy. Past data can be recast as per NCO 04 with the help of **concordance tables provided in NCO 04**.

A word of caution is necessary while using data assembled from sources other than one's own. Data extracted from printed publications or websites have to be checked for printing errors/errors that occur while data are being posted on the websites. A check of the row and column totals of the tables and other identity relations relevant to the data under reference in the tables should take care of such errors. More importantly, you should also look into aspects of the data, like the concepts, definitions and methodology adopted by the source-agency for collecting and compiling the data, coverage of data (like, whether data was collected from entire country or some specific areas, whether the sample drawn was based on a statistically reliable random sample or it was a purposive sample of a few selected locations, etc.), the methods used for collecting data, reliability and so on periodicity and timeliness in the availability of data, integrity (confidentiality), the data are firmly moored to what the data are actually supposed to reflect. These aspects are collectively referred to as “**metadata**”. All these details are generally available in the publication containing the data or in a related publication of the agency publishing the data or in the relevant website of the agency.

It would sometimes be useful and instructive to judge the Indian situation in an international setting. Publications of the United Nations and its agencies as also

agencies like the International Monetary Fund (IMF) and the World Bank and regional agencies such as the Organisation for Economic Cooperation and Development (OECD)¹ provide data on different facets of the economies of the member countries including India along with information regarding comparability of country-wise data. The IMF has formulated a “**Special Data Dissemination Standards**” (SDDS) covering the real sector (national accounts, production index, price indices, etc.), Fiscal Sector, Financial Sector, External Sector and Socio-demographic data for different countries to facilitate transparency in the compilation and dissemination of data on important aspects of the economy of individual countries and cross-country comparison of such data. Countries like India who have accepted the SDDS provide to **IMF** a **National Summary Data Page** in respect of each of the areas and sub-areas listed in SDDS and metadata relating to such data, as per a Dissemination Format prescribed in SDDS. In addition, individual country agencies concerned disseminate an advance release calendar, which gives notice of the precise dates of release of data three months ahead of the date of release of the data, on the internet of the IMF’s Data Dissemination Bulletin Board (DSBB). Such information, as provided by any country covered by SDDS, can be accessed on the internet with the help of the Google search engine using the search parameter Special Data Dissemination Standards IMF. You can also visit the site of the United Nation’s Statistical Division databases at <http://unstats.un.org/unsd/databases.htm> to access data compiled by different UN agencies on a host of economic and social parameters.

Your capacity to utilize the vast amount of data available in the Indian Statistical System and examine critically the state of the society can be enhanced considerably by an intelligent combination of data across data domains. You have already come across some examples of such efforts – poverty ratios, concentration ratios for measuring the level of inequality, employment elasticity, labour productivity and capital productivity. (Unit 4 on Poverty and Inequality – Policy Implications and Unit 5 on Employment and Unemployment – Policy Implications in Block 1 of MEC-005 on Indian Economic Policy in your first year Course). Other examples are the Human Development Index and Gender Development Index (Units 3 and 4 in the same Block of MEC-005).

Check Your Progress 1

- 1) State the major functions of CSO.

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¹ India is not a member of OECD, which is an organization in which several countries in Europe are members.

- 2) Name any three web sources where you can get data for your research, free of cost.

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- 3) Which precautions should be taken care of while using data assembled from various sources?

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- 4) What do you understand by the term Special Data Dissemination Standard?

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Let us now turn our attention to data on macroeconomic variables like national income, saving and investment.

21.3 NATIONAL INCOME AND RELATED MACRO ECONOMIC AGGREGATES

21.3.1 System of National Accounts (SNA)

How do you assess the performance of an economy? You may be able to look at the trend in the production of rice or wheat to be able to say something about the performance of paddy or wheat crop. You can make a similar assessment about the production of steel. If you want to say something on agricultural production, where you find that some crops have done well and others have not, you think of making an overall assessment of agricultural performance by constructing an index of agricultural production to review the performance of agricultural production. But we should like to go beyond levels of output or production and look at performance in terms of incomes flowing from output in the form of rent, wages, interest and profit to those participating in the creation of output namely the factors of production – land, labour, capital and entrepreneurship (Since 1993, the role of the Government in influencing the economy, through various taxes and subsidies have also become an important item of analysis). Alternatively, we would like to base our judgment of

performance on value addition made by the production system namely, value of output net of the (intermediate) costs incurred in creating the output. It is (i) this overall value addition computed for all sectors/activities of the economy, that is referred to as the National Product, (ii) macro-aggregates related to it, and (iii) trends in (i) and (ii), that can help you in analyzing the performance of an economy.

As you know, National Income (NI) is the Net National Income (NNI). It is also used to refer to the group of macroeconomic aggregates like Gross National Income (GNI), Gross Domestic Product (GDP), Gross Value Added (GVA) and Net Value Added (NVA). All these of course refer to the total value (in the sense mentioned above) of the goods and services produced during a period of time, the only differences between these aggregates being depreciation and/or net factor income from abroad. There are other macroeconomic aggregates related to these that are of importance in relation to an economy. What data would you, as an analyst, like to have about the health of an economy? Besides a measure of the National Income every year or at smaller intervals of time, you would like to know how fast it is growing over time. What are the shares of the national income that flow to labour and other factors of production? How much of the national income goes to current consumption, how much to saving and how much to building up the capital needed to facilitate future economic growth? What is the role of the different sectors and economic activities – in the public and private sectors or in the organized and unorganized activities or the households in the processes that lead to economic growth? How does the level and pattern of economic growth affect or benefit different sections of society? How much money remains in the hands of the households for consumption and saving after they have paid their taxes (Personal Disposable Income) – an important indicator of the economic health of households? What is the contribution of different institutions to saving? How is capital formation financed? Such a list of requirements of data for analyzing trends in the magnitude and quality of, and also the prospects of, efforts for economic expansion being mounted by a nation can be very long. Such data, that is, estimates of national income and related macroeconomic aggregates from part of a system of National Accounts that gives a comprehensive view of the internal and external transactions of an economy over a period, say, a financial year and the interrelationships among the macroeconomic aggregates. National Accounts thus constitute an important tool of analysis for judging the performance of an economy vis-à-vis the aims of economic and development policy.

The United Nations (UN) has been recommending guidelines in the form of a System of National Accounts (SNA) in order to promote international standards for the compilation of national accounts as an analytical tool and international reporting of comparable national accounting data. The first version of SNA was recommended in 1953. This was revised in 1968 and then in 1993. The fourth version (2008 SNA) was prepared under the auspices of the Inter-Secretariat Working Group on National Accounts consisting of the Commission of the European Communities (Eurostat), International Monetary Fund (IMF), Organisation for Economic Cooperation and Development (OECD), the UN and the World Bank and adopted by the UN Statistical Commission in 2008. This system has been harmonized with other major statistical systems like the **Balance of Payments Statistics and Government Finance Statistics of the IMF**. The **2008 SNA** contains a coherent, consistent and integrated set of macroeconomic accounts based on a set of internationally agreed concepts, definitions, classifications and accounting rules. It provides a

comprehensive accounting framework within which data can be compiled and presented in a format that is designed to facilitate economic analysis, formulation of policy and decision-making. You can read the entire SNA by accessing it at <http://unstats.un.org/unsd/nationalaccount/docs/SNA2008.pdf>.

The informal sector or the unorganized sector occupies an important place in our economy, in terms of contribution to national income, employment and exports. It is necessary, therefore, to bestow adequate attention to unorganized sector in the compilation of National Accounts and in other data domains and in international efforts at promoting collection of data on informal sector activities. The 15th International Conference of Labour Statisticians (ICLS) (January, 1993) adopted a resolution on informal sector statistics with a view to helping member countries of the International Labour Organisation (ILO) in reporting comparable statistics of employment in the informal sector. The resolution was endorsed by the UN Statistical Commission and by 1993 SNA.

National Accounts are compiled in India using a mix of 1968 SNA and 1993 SNA. India is also in the process of moving towards the full implementation of SNA methodology to the extent feasible.

21.3.2 Estimates of National Income and Related Macroeconomic Aggregates

a) Estimates Released by CSO

The Central Statistics Office (CSO) in the Ministry of Statistics and Programme Implementation (MoSPI), Government of India compile and publish National Accounts, which include estimates of National Income and related macroeconomic aggregates like NNP, GNP, GDP and NDP, PFCE (private final consumption expenditure), saving, capital formation and so on for the country and for the public sector for every financial year. Quarterly estimates of GDP are also prepared and released. Estimates prepared for any year at the prices prevailing in that year are called estimates *at current prices*. Estimates of national income and other aggregates at current prices is generally not used in analyzing changes in the magnitude of these aggregates over time, as price levels over time gets affected by inflation or deflation of an economy. Suppose we need to compare the performance of the economy in terms of national income or other macroeconomic aggregates, over a period of time, say, five years. A comparison of estimates of, say, national income at *current prices* in the opening year and the final year of the five-year period will give us the increase or decrease, as the case may be, in national income at current prices. We can easily note that the quantum of change observed in national income is *the composite measure of the change in national income and the changes in the prices of goods and services between the two points of time*. How then to get at the actual change, or the change in real terms in national income (or any other macroeconomic aggregate) over the period? This is done by removing the effect of changes in prices while comparing the aggregate in question at different points of time. Estimates of the aggregate for different years are, therefore, prepared at the estimates of the aggregate at constant (base year) prices. The comparison of estimates of the aggregate at constant (base year) prices at different points of time is comparison of the magnitude of the aggregate in real terms and measures the real change over time in the magnitude of the aggregate. The base year now in use is 2011-12.

CSO changes the base year from time to time in order to take into account the structural changes that take place in the economy and depict a true picture of the economy. It was 1948-49, initially, when estimates of National Income were first prepared by CSO and published in 1956. It was shifted to 1960-61 in August, 1967, to 1970-71 in January, 1978, to 1980-81 in February, 1988, to 1993-94 in February, 1999 to 1999-2000 in January, 2006; to 2004-05 in January 2010 and to 2011-12 in January 2015. Note that, in the beginning, the base years were the years in which the decennial population census was conducted, that is, the first year of every decade from 1961 to 1981. The choice of the subsequent base years, namely, 1993-94, 1999-00 and 2004-05 were a departure from this practice. What are the reasons for this? Estimates of workforce participation rates (WPR), that is, the proportion of workers to population for the benchmark years play an important role in the compilation of estimates of national income. CSO had been making use of data on WPR available from decennial population censuses till 1981 while compiling national income estimates and for this reason the base year was the relevant decennial census year. It was, however, observed that the Employment and Unemployment Surveys of the National Sample Survey Organisation (NSSO) captured WPR better than the population censuses. The base years chosen after 1981 were thus 1993-94, 1999-2000 and 2004-05, making use of data on employment based on the NSSO 50th, 55th, 61st Round and 68th round Surveys on Employment and Unemployment² conducted during July '93 – June '94, July '99 – June 2000, July '03 – June '04 and July 2011- June 2012 respectively. After every revision of base year, the CSO publishes a set of results which provide comparable estimates of the years prior to the base year, so that a long time series data on national income aggregates can be analysed by the researchers and policy makers.

Base year revisions differ from annual revisions in National Accounts primarily because of nature of changes. In annual revisions, changes are made only on the basis of updated data becoming available without making any changes in the conceptual framework or using any new data source, to ensure strict comparison over years. In case of base year revisions, apart from a shift in the reference year for measuring the real growth, conceptual changes, as recommended by the international guidelines, are incorporated. Further, statistical changes like revisions in the methodology of compilation, adoption of latest classification systems, and, inclusion of new and recent data sources are also made. Changes are also made in the presentation of estimates to improve ease of understanding for analysis and facilitate international comparability.

Estimates of national accounts aggregates for different years are published in considerable detail in CSO's Annual Publication "National Accounts Statistics (NAS)". The latest available NAS estimates are detailed in the publication NAS 2014 and the Press Notes of MoSPI dated the 31st January, 2015³. The press note of January 2015 provides, for the first time, estimates of macro-economic aggregates using the base year 2011-12. Note that in this release, for the first time, Gross Domestic Product (GDP) "at factor cost" has been discontinued. As is the practice internationally, industry-wise estimates have been presented as Gross Value Added (GVA) at basic prices, while "GDP at market prices has been referred to as GDP. Estimates of GVA at factor cost

² Some employment data from Census 1991/2001 have also been used to supplement the data from NSSO.

³ This publication and these Press Notes can also be accessed in the **Ministry of Statistics and Programme Implementation website** www.mospi.nic.in